Temporal Convolutional Network (TCN) for RoNIN Heading Estimation

Model Report: Additional Baseline

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Field	Value
Model Name	Temporal Convolutional Network (TCN)
Dataset	RoNIN preprocessed split
Input	200 x 9 sensor window
Output	sin(heading), cos(heading)

1. Architecture

1.1 Model Description

This model estimates pedestrian heading from RoNIN smartphone sensor windows using a Temporal Convolutional Network. Each input sample contains one second of synchronized accelerometer, gyroscope, and magnetometer readings with shape 200 x 9. The model predicts sin(heading) and cos(heading), which are converted back to degrees during evaluation.

1.2 Layer Structure

Component	Details
Input	200 x 9 sensor window
TCN block 1	Conv1D, 9 -> 64 channels, kernel size 3, dilation 1, ReLU, dropout 0.2, residual projection
TCN block 2	Conv1D, 64 -> 64 channels, kernel size 3, dilation 2, ReLU, dropout 0.2, residual connection
TCN block 3	Conv1D, 64 -> 64 channels, kernel size 3, dilation 4, ReLU, dropout 0.2, residual connection
TCN block 4	Conv1D, 64 -> 64 channels, kernel size 3, dilation 8, ReLU, dropout 0.2, residual connection
Sequence summary	Final timestep output only
Fully connected output	64 -> 2 output neurons

1.3 Trainable Parameters

Total trainable parameters: 39,618.

1.4 Architectural Choices and Justification

The TCN was selected because it provides a learned temporal baseline between a flat MLP and recurrent or attention-based models. Dilated causal convolutions allow the network to capture short and medium-range temporal patterns while avoiding recurrent computation. Four dilation levels, 1, 2, 4, and 8, were used to gradually expand the temporal receptive field. A channel size of 64 kept the model compact compared with the LSTM, Transformer, and Hybrid models, while dropout of 0.2 was used to reduce overfitting. Residual connections were included to support gradient flow through the convolutional stack. This design follows the motivation of Bai et al. (2018), where TCNs are presented as strong sequence modelling alternatives to recurrent networks.

2. Training Setup

2.1 Loss Function

Mean Squared Error (MSE) loss was applied to the two predicted outputs, sin(heading) and cos(heading). This avoids the discontinuity problem that occurs when heading is predicted directly in degrees, especially around the 0/360 degree boundary. During evaluation, the sine and cosine outputs were converted back to heading degrees using arctan2, and circular heading error was used.

2.2 Training Hyperparameters

Hyperparameter	Value
Optimizer	Adam
Learning rate	0.001
Batch size	64
Maximum epochs	50
Early stopping	Validation loss, patience 5
Device used	Apple MPS

2.3 Learning Rate Scheduling

None used. The learning rate was kept fixed at 0.001 to match the fixed settings for the TCN baseline.

2.4 Early Stopping and Timing

Early stopping monitored validation loss with patience 5. The best validation loss was 0.407983 at epoch 28, and training stopped at epoch 33 after validation loss stopped improving. The recorded training time was 37,256.36 seconds, which is approximately 620.94 minutes. This is longer than expected for a TCN, and may be affected by Apple MPS overhead, full validation at every epoch, notebook runtime overhead, thermal throttling, or idle/sleep time during execution.

2.5 Training Difficulties

The main difficulty was computational time. Although TCNs are usually expected to train faster than recurrent and attention-based models, this run took longer than the other reported models. The validation loss also fluctuated slightly while the training loss continued to decrease, suggesting that generalisation to unseen subjects remained challenging.

3. Results

3.1 Quantitative Results

Metric	Value
MAE on test set (degrees)	69.2654
RMSE on test set (degrees)	85.9966
Best validation loss	0.407983 at epoch 28
Approximate training time	37,256.36 s / 620.94 min
Trainable parameters	39,618

3.2 Predicted vs Ground Truth Trajectory

The plot below shows the first 1,000 unseen test windows. The model captures some broad stable regions, but it also shows large deviations and abrupt jumps in dynamic sections. This supports the relatively high MAE on the unseen-subject test set.

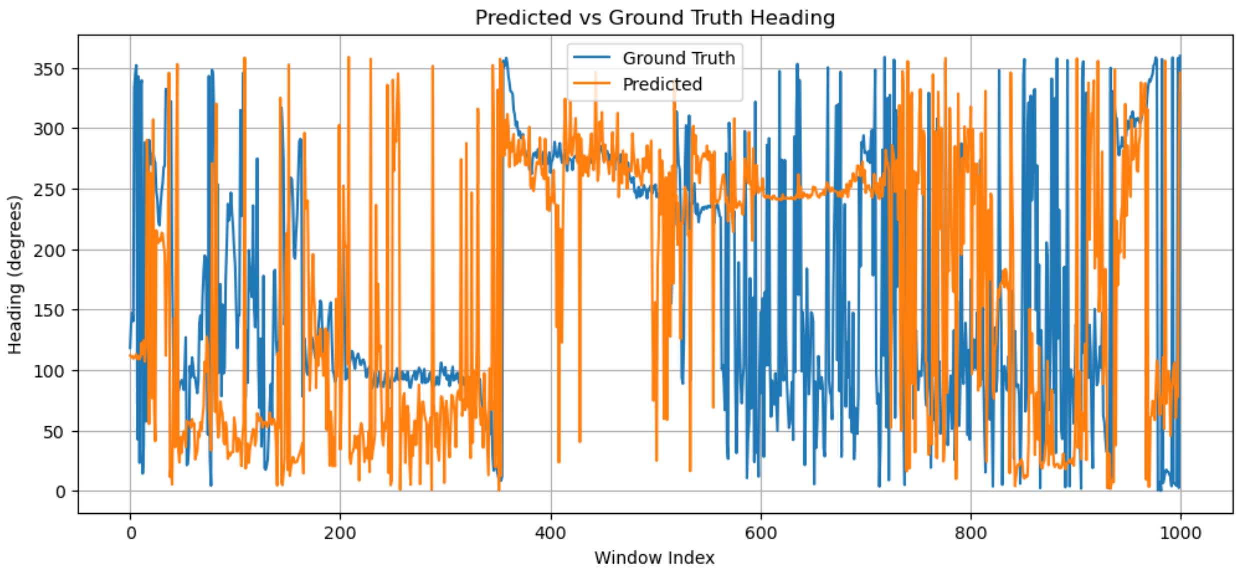

Figure 1. Predicted vs ground truth heading on unseen test windows.

3.3 Loss Curve

The training loss decreases consistently, while the validation loss improves more slowly and begins to flatten after the middle of training. The best validation result occurs at epoch 28. The increasing gap between training and validation loss suggests mild overfitting, but not complete training failure.

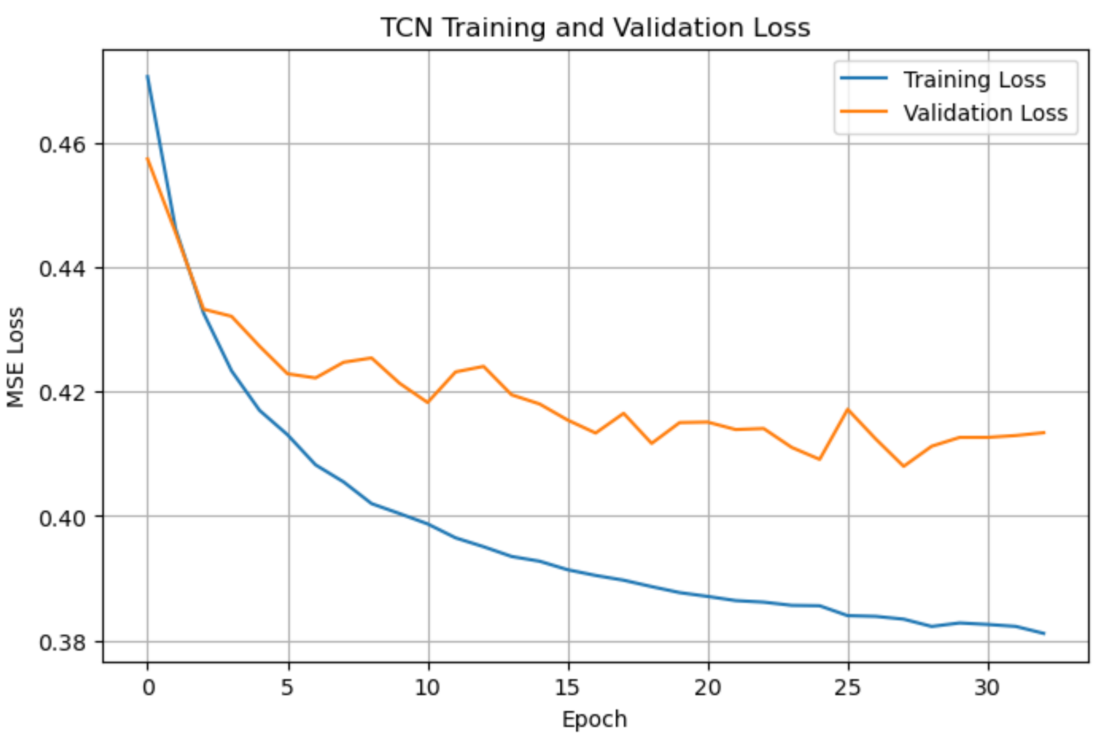

Figure 2. TCN training and validation loss.

3.4 MAE Breakdown by Scenario

Scenario	MAE (degrees)
Straight walking	63.3921
Sharp turns	87.8784
Short trajectories	74.3783
Long trajectories	64.1524

Scenario-wise MAE was computed using circular heading error. Straight walking produced the lowest error at 63.3921 degrees, while sharp turns produced the highest error at 87.8784 degrees. Short trajectories had 74.3783 degrees MAE and long trajectories had 64.1524 degrees MAE. This shows that the TCN handles smoother and longer stable motion better than abrupt or short motion patterns.

4. Observations and Technical Reflection

4.1 Where the Model Performs Well

The model performs best in smoother parts of the trajectory where heading changes gradually. This is supported by the scenario-wise results, where straight walking has the lowest MAE of 63.3921 degrees. In these sections, local temporal patterns in acceleration, gyroscope, and magnetometer readings are more consistent, allowing the convolutional filters to capture useful motion structure. The lower error on long trajectories, 64.1524 degrees, also suggests that the TCN can remain reasonably stable when the motion pattern is less abrupt.

4.2 Where the Model Struggles

The model struggles most during abrupt heading changes and short or highly variable motion. Sharp turns have the highest MAE at 87.8784 degrees, showing that rapid directional changes are difficult for the convolutional architecture to capture accurately. Short trajectories also have a relatively high MAE of 74.3783 degrees. This is expected because the model only receives a one-second window and uses the final timestep representation, so sudden transitions within the window may not be represented strongly enough.

4.3 Loss Curve Interpretation

The loss curve shows that the model learns from the training data, as training loss decreases from 0.470667 to 0.381115. Validation loss also improves overall, reaching its best value of 0.407983 at epoch 28, but the improvement is slower and more unstable. This indicates that the model converges, but generalisation to unseen subjects remains limited. The pattern suggests mild overfitting rather than underfitting.

4.4 Architecture vs Training; Root Cause of Limitations

The limitations likely come from both the architecture and the training setup. Architecturally, a TCN uses fixed convolutional filters over a limited temporal receptive field. This is efficient, but it may be less flexible than recurrent gating or attention when heading changes quickly. The scenario results support this: performance is better for straight walking and long trajectories, but much worse for sharp turns. From the training side, the model is evaluated on unseen subjects, so differences in walking style, phone placement, and sensor noise make generalisation difficult. The long training time also limited the opportunity for further tuning.

4.5 Further Improvement

If more time was available, I would tune the TCN with a smaller and faster configuration first, then compare channel sizes such as 32, 64, and 128 with dropout values such as 0.1, 0.2, and 0.3. I would also add a learning rate scheduler such as ReduceLROnPlateau to improve validation convergence and reduce unnecessary training time.

5. Cross-Model Comparison

Model	MAE (degrees)	RMSE (degrees)	Comment
Kalman Filter	86.79	102.13	Weakest overall, no learning
LSTM	67.05	83.61	Better than TCN by about 2.22 MAE
Transformer	66.99	83.30	Better than TCN by about 2.28 MAE
Hybrid LSTM-Transformer	66.1114	82.8781	Best of the listed models
TCN	69.2654	85.9966	Clearly better than Kalman, slightly weaker than the other learned models

Among the provided results, the TCN ranks above the Kalman Filter but below the LSTM, Transformer, and Hybrid models. This is still a useful result because it shows that convolutional temporal modelling improves over a traditional filter-based baseline but does not fully match recurrent or attention-based temporal modelling on this setup. MLP results were not provided, so a direct numerical comparison against MLP cannot be made.

6. Summary

Overall, the TCN baseline is complete and comparable to the other learned models. It achieves a test MAE of 69.2654 degrees and RMSE of 85.9966 degrees on the unseen-subject test set. The scenario-wise results show its clearest weakness on sharp turns, with 87.8784 degrees MAE, while straight walking is more stable at 63.3921 degrees MAE. Although it does not outperform the LSTM, Transformer, or Hybrid models, it performs much better than the Kalman Filter and adds a useful convolutional temporal baseline to the group comparison.

References

Bai, S., Kolter, J. Z., & Koltun, V. (2018). An empirical evaluation of generic convolutional and recurrent networks for sequence modeling. arXiv:1803.01271.

Herath, S., Yan, H., & Furukawa, Y. (2020). RoNIN: Robust neural inertial navigation in the wild: Benchmark, evaluations, & new methods. IEEE International Conference on Robotics and Automation (ICRA), 3146-3152.